

Circular Capital in the Artificial Intelligence Stack: A Game-Theoretic Analysis of Market Power, Financial Fragility, and Welfare

Anonymous for double-blind review

Abstract

The 2024–2026 artificial-intelligence investment boom shows the financing structure behind modern round-trip busts: dense, committed circular capital in which upstream investment returns as contracted compute spend, measured at \$675.40 billion across 17 committed edges. It concentrates burst incidence upstream. Bubble conditions score $\sim 83\%$ at frontier labs (Foundation Models), where private valuations detached furthest from capturable revenue. Credit markets now charge for stack leverage (single-name CDS at a record 215bp; a BBB-downgrade). Six solved bilateral games yield aggregate deadweight loss of \$416.77 billion at 71.28% efficiency, with non-cooperation binding in the Hardware–Cloud game. A severe burst destroys $\sim \$239$ billion of Hardware value and $\sim \$133$ billion of Cloud value first, costing the US economy $\sim \$139$ billion ($\sim 0.45\%$ of GDP) through capex and wealth channels. The chain is mechanical: signed circular contracts trap value in the Hardware–Cloud game, the trap fixes the loss order, and the order survives 16 extremes and 300 joint draws. Run blind on 2008 inputs, the same modules recover the historical loss order (thirteen of fourteen verdicts). The dollar figures size remedies; the ranking is the forecast. By benefit–cost ratio, disclosure and lending caps come first. Three stated observables would retire the call.

Keywords: artificial intelligence; financial bubbles; circular dealing; vendor financing; game theory; financial stability.

JEL codes: D43, G01, G12, G14, L13, L41, E44, O33.

1 Introduction

This paper converts a measured financing structure into a dated, testable prediction: an AI-asset repricing is likely while the current committed spend cycle runs; its losses will land first on Hardware, then Cloud, then frontier labs, then apps; and the severe case costs the US economy roughly \$100–150 billion. Section 2 states that forecast as eight numbered claims, each with an assessed likelihood (or an explicit abstention, P8) and a pointer to the observable that would overturn it. The remaining sections supply the evidence (Sections 5–11), the record of stress testing and historical validation (Sections 12–13), and the investment and policy implications (Sections 14–16).

The argument is a five-link chain. **Measurement:** \$675.40 billion of committed circular capital read off signed contracts (Section 5). **Mechanism:** six solved games show non-cooperation binds in exactly one cell — Hardware versus Cloud (Sections 6–7). **Incidence:** the trap fixes the burst order, with formation worst where private marks detached furthest (Sections 9–10). **Validation:** the order survives sixteen extremes and 300 joint draws, and the same modules recover the 2008 loss order and the 2001 telecom mechanism (Sections 12–13). **Action:** triggers plus remedies sequenced by benefit–cost ratio (Sections 14–16). Throughout, *rankings* are the high-confidence claims; *dollar figures* mark scenario scale for sizing remedies.

2 Central Forecast: Repricing Likelihood, Loss Incidence, and Macroeconomic Cost

This paper makes a prediction with three parts. **First**, an AI-asset repricing is likely while the current committed spend cycle runs: the conditions that have preceded every modern round-trip bust — dense reciprocal financing, private marks detached from capturable revenue, and credit markets starting to charge for leverage — are all present at once. **Second**, the damage has a fixed address: Hardware first (~\$239B), Cloud second (~\$133B), frontier labs third (~\$85B), app companies last (~\$21B), because contracts, not sentiment, decide where losses land. **Third**, the US GDP cost of the severe case is ~\$139B (~0.45%), composed of canceled tech investment (~\$120B) plus the wealth effect of falling equities (~\$19B).

What this paper does *not* predict is a date. Timing belongs to funding rounds, credit events, and capex guidance — the triggers of Section 14 — not to models. Everything else here is a measured claim, not a scenario sketch: the capital structure is read off contracts, the binding constraint is proved in solved games, and the resulting incidence ranking is the same order that survives every stress battery and transfers to 2008. The result is a standing warning with an address book: it tells you what breaks, in what order, at roughly what scale, and what to do before it happens.

How seriously to take each claim. Each prediction carries a **likelihood** (author-assessed probability on the fixed scale in Table 1) and a **confidence** grade (quality of evidence). Model vote-shares measure *robustness* and formation scores *vulnerability* — neither is an event probability.

Table 1. Likelihood scale for author-assessed probabilities. Used only in the registry below; nowhere else in the paper does a percent sign denote an event probability.

Wording	Assessed probability
Very likely	85–95%
Likely	65–80%
More likely than not	50–65%
Toss-up	~50%
Unlikely	below 35%

Read P2 as written: *conditional* — ~90% on the ranking *given* a severe burst, whose occurrence is P1’s ~65%. Jointly “a severe burst with Hardware first” sits near 60%: a standing warning with an address book, not a forecast.

Table 2. Prediction registry: every claim, its likelihood, its evidence grade, and what retires it.

ID	Claim	Likelihood	Confidence	Overtured by
P1	AI-asset repricing before the committed spend cycle exhausts itself	Likely ($\sim 65\%$)	Medium	Validation tests 1–2
P2	<i>Conditional on a severe burst</i> , losses rank Hardware > Cloud > labs > apps	Very likely ($\sim 90\%$)	High	Test 3 (ranking flips)
P3	Severe GDP cost lands near \$100–150B (~ 0.3 – 0.5%), mostly via canceled capex	More likely than not ($\sim 55\%$)	Medium	A burst with no capex cut
P4	Frontier-lab funding stress is the nearest trigger	Likely ($\sim 70\%$)	Medium	A first break elsewhere (Sec. 14)
P5	Labs carry the highest bubble vulnerability (formation rank FM first)	More likely than not ($\sim 60\%$)	Medium	Formation rank flips (65% joint agreement)
P6	Round two is small ($\sim \$15$ B), mostly on labs ($\sim \$13.9$ B of $\sim \$15.2$ B)	More likely than not ($\sim 60\%$)	Medium	Collateral fire sale (Sec. 11)
P7	Disclosure first, lending caps second is the optimal remedy sequence (caps still rank first by BCR)	Likely ($\sim 70\%$)	Medium	BCR rank reversal (Sec. 16)
P8	Any calendar date for the repricing	— (unassigned)	Low	— (no date predicted)

3 Analytical Approach: Contractual Exposure as the Forecasting Primitive

Most bubble calls argue about prices. This one argues about *plumbing*: the \$675.40B of committed round-trip financing, across 17 booked deals, that wires the AI stack into a single fate-sharing machine. When company A invests in company B and B contractually spends the money back at A, two balance sheets become one balance sheet wearing two logos. Prices can then stay high for good reasons (real revenues: NVIDIA’s record \$57.0B quarter, Q3 FY2026) while the financing underneath stays fragile. That split — real revenues, fragile financing — is what makes a burst prediction possible without a price prediction: you do not need to know when enthusiasm fades, only where the contracts point when it does. The machinery is a single Python pipeline of 35 classes covering data validation, market structure, solved games, welfare accounting, bubble diagnostics, and valuation, gated end to end by 364 tests.

Why this toolkit and not price charts. Traditional bubble models study anonymous markets through prices, sentiment, or aggregates; the AI stack is the opposite — a handful of giants negotiating bespoke bilateral contracts — so the tools model the plumbing directly. *Game theory* solves each pairwise negotiation for its stable outcome, exposing the one gridlocked cell. The *Herfindahl–Hirschman Index* (the DOJ concentration screen, highly concentrated above 2,500) quantifies the chokepoint: at 3,851, no diversified fringe exists to absorb a failure. *Saltelli global sensitivity analysis* (the variance-decomposition workhorse of physics and climate modeling) then checks which input actually drives fragility: shared circular capital scores a total-order index of 0.99, confirming the prediction emphasizes the variable the system is most sensitive to.

4 Literature Review: Bubbles, Platforms, and Fragility

This prediction does not start from zero. Five strands of published research supply its tools, its benchmarks, and its warnings; this section states each in one paragraph, with pointers for readers who want the originals. Full references sit in the References list, in APA style.

1. *How economists spot bubbles.* The standard statistical toolkit hunts explosive price paths: Phillips et al. (2011) built the SADF test that date-stamps when exuberance starts and collapses, and Phillips et al. (2015) generalized it to multiple bubbles. The dot-com precedent — world-changing technology alongside prices far above eventual cash flows — is documented by Ofek and Richardson (2003), while Greenwood et al. (2019) show such run-ups predictably end in crashes. Takeaway for this paper: price tests tell you *whether* something looks explosive; they cannot tell you *where the losses land*. The financing graph modeled here answers the second question, which is why the two approaches complement rather than compete.

2. *What AI does to the economy.* The economics-of-AI program (Agrawal et al., 2018, 2019) now measures real generative-AI productivity gains (Brynjolfsson et al., 2025; Noy and Zhang, 2023) against slow general-purpose-technology payoffs (Brynjolfsson et al., 2021; David, 1990), while macro watchdogs flag AI-correction risk, big-tech leverage, and supply-chain funding complexity (IMF, 2026b; Adrian, 2026; Bank for International Settlements, 2026). Takeaway: the revolution is real *and* the financing is fragile.

3. *Why platforms lock customers in.* Cloud bundling follows the information-goods logic of Shapiro and Varian (1999) and Bakos and Brynjolfsson (2000); switching costs and network effects are surveyed by Farrell and Klemperer (2007); sunk-cost market structure by Sutton (1991); semiconductor competition by Scherer and Ross (1990). On policy, the Digital Markets Act (European Commission, 2022), the digital-era competition report (Cr  mer et al., 2019), vertical-stack analysis (Hagiu and Wright, 2025), the antitrust-paradigm literature (Baker,

2019), and data-portability economics (Krämer et al., 2020) shape the remedy menu, while the FTC documents investment capital bundled with cloud spend (Federal Trade Commission, 2025). Takeaway: lock-in theory predicts exactly the \$142.62B switching-cost envelope this paper measures — the oldest literature here correctly calls the largest waste category. The classic monopoly-waste benchmark is Harberger (1954), and the reason fixing one broken margin can backfire elsewhere is Lipsey and Lancaster (1956).

4. *Vendor financing and fire sales.* Round-trip financing has a known pathology (telecom vendors lending to their own buyers), and the fire-sale mechanic — homogeneous collateral worth less together than apart — is formalized by Shleifer and Vishny (1992). The cascade engine in this paper is the DebtRank method of Battiston et al. (2012). Takeaway: every moving part of the contagion story is borrowed from established work; the novelty is wiring them into one stack with measured exposures.

5. *Concentration and hype measurement.* AI markets concentrate through fixed training costs and size advantages even without misconduct (Korinek and Vipra, 2025); scale, GDP, and valuation anchors come from institutional-grade public sources (IEA, 2026; International Data Corporation, 2026; PwC, 2026; U.S. Bureau of Economic Analysis, 2026; Congressional Budget Office, 2022; Damodaran, 2026), listed to the row in the pipeline’s 110-row register.

5 Measurement of Circular Capital: Scale and Composition

Seventeen committed deals, each traced to primary reporting and re-verified in September 2026, put \$675.40B of circular capital on the books (\$1,188.33B tracked including two memo rows); the largest rows are a \$249.20B aggregate GPU-supply exposure and a \$105B NVIDIA credit guarantee (Table 3). Parked *outside* the booked total, exactly as they should be: Google’s \$30B conditional tranche, Amazon’s \$20B, an NVIDIA tranche of up to \$10B, decade-horizon reverse spends, the \$500B six-platform MOU facility (aspiration, not a contract), and the \$12.93B NVIDIA–Hugging Face acquisition (a purchase price for control, with no reverse-spend leg).

Table 3. Largest committed circular exposures (\$B). Full 19-row frame in the pipeline’s validated deal ledger (pipeline Table 3 series).

Exposure	Committed
Aggregate GPU-supply exposure	249.20
NVIDIA \$105B Ohio credit guarantee	105.00
Lambda + Nscale compute commitment	80.00
Fluidstack compute	50.00
SpaceX Colossus	45.00
AWS cloud agreement (OpenAI)	38.00
Azure purchase (Anthropic)	30.00
CoreWeave contract (Meta)	14.20
Microsoft–OpenAI / Amazon–Anthropic	13.00 each
Committed total (17 edges)	675.40
Tracked total (19 rows)	1,188.33

Prediction consequence: one row (GPU supply, 36.9%) and one pair (Hardware–Cloud) dominate the frame, and Hardware (54.7%) plus Cloud (42.2%) own 97% of the 3,851-point concentration index — so the burst has a chokepoint. More than three-quarters of the frame is procurement (propagating at face in DebtRank), while the 15.5% guarantee layer is contingent: small in the base case, outsized in the severe one.

6 Strategic Interaction: Six Bilateral Games and the Scope for Cooperation

The core game is the prediction’s engine, but the other five set its boundaries. Each row below gives the stable outcome, the best-joint outcome, the trapped value, and the payment that would unlock it.

Table 4. All six games: stable vs. best-joint welfare (\$B).

Game	Nash	Joint max	Trapped	Transfer to unlock
HW–Cloud	1,094.8	1,153.4	58.6	176.0 (HW to Cloud)
HW–Foundation	671.3	876.9	205.6	46.7 (HW to FM)
HW–Wrappers	609.0	856.1	247.0	18.7 (HW to Wrappers)
Cloud–Foundation	604.1	745.4	141.2	46.3 (Cloud to FM)
Cloud–Wrappers	544.3	725.0	180.7	18.7 (Cloud to Wrappers)
FM–Wrappers	142.5	177.1	34.6	1.1 (FM to Wrappers)
Diagnostic sum			867.8	307.4

Three prediction readings. *First*, the largest trapped value (\$247.0B, Hardware–Wrappers) needs the smallest key (\$18.7B): the cheapest cooperation in the stack is also among the most valuable, which is where a negotiated settlement would start. *Second*, every unlock payment flows upstream-to-downstream, so in a crisis the kill chain runs through whoever refuses to pay — watch Hardware’s and Cloud’s willingness to compensate, not the labs’. *Third*, the core game’s \$176.0B price tag on a \$58.6B gain is the worst ratio in the stack: the most important cooperation is the least likely to happen voluntarily, which is why it needs regulation rather than hope. (The \$867.8B sum multi-counts shared players across pairs; the \$416.77B top-down bound, not this sum, sizes policy.)

Stability sharpens the timing logic (Table 7.4): four of six games are stable loss pockets ($\geq 80\%$ Nash reproduction), while the core game reproduces at 64.7% — the modal outcome, not a certainty. P2’s $\sim 90\%$ burst ranking follows contracts rather than equilibria, so even draws where the core game flips leave the loss address unmoved.

7 The Binding Constraint: Non-Cooperation in the Hardware–Cloud Game

Six pairwise standoffs were solved. All six trap value absent cooperation, but only one binds the prediction: Hardware vs. Cloud. Chipmakers stay open while clouds stay locked in — neither side wants to move alone — trapping \$58.6B that cooperation would unlock (94.9% efficient). That single stuck relationship, sitting on the largest contract base in the stack, is the structural reason a repricing becomes a loss rather than a shrug.

Table 5. The core Hardware–Cloud game (\$B). The stuck cell is the selected equilibrium; the best-joint cell is Nash too — payoff- and risk-dominant — but unselected, so persistence of the inferior cell is a lock-in story, not an incentives story.

	Cloud: Open	Cloud: Locked (Nash)
Hardware: Open (Nash)	—	487.9 + 606.8 = 1,094.8
Hardware: Walled (Pareto)	722.5 + 430.8 = 1,153.4	—

Two audits sharpen the reading: the efficient outcome is payoff- and risk-dominant (Table 3.6) yet unselected — lock-in, not incentives, holds the inferior cell — and cooperation is sustainable without transfers in three of six games including the core, while the three lab games cannot sustain it at any patience level (Table 3.7).

8 Welfare Analysis: Decomposition of Deadweight Loss

The stuck games waste \$416.77B in total, itemized in Table 6. The prediction-relevant fact is the ordering, which survives every stress test: lock-in first (\$142.62B), distorted R&D second (\$106.97B). Remedies aimed anywhere else are aimed at smaller problems.

Table 6. Waste by source (\$B). Order first, levels second.

Source	Loss
Switching costs (lock-in)	142.62
Innovation distortion	106.97
Quality degradation	71.31
Coordination failures	60.01
Monopoly pricing	35.86
Total	416.77

9 Bubble Vulnerability Assessment by Archetype

Bubble-condition scores (vulnerability ratings from a scenario-calibrated model, not measured probabilities; bands plotted below, drivers in the `pred_E` ledger) peak exactly where round-tripping is densest.

Prediction consequence: monitor frontier-lab funding rounds as the leading indicator. Each mega-round at a higher valuation re-prices every adjacent private mark at once. Anthropic at \$965B, Databricks near 905x earnings, and xAI at 400x revenue have the furthest to fall.

Table 7. Bubble-score driver decomposition (logit inputs; Table 7.10). Circular exposure dominates by construction (weight 1.6); exuberance alone cannot lift a score.

Archetype	Circ.	Capex	HHI sh.	Exub.	Score
Foundation Models	0.80	0.70	0.10	1.00	82.9
Hardware	0.40	0.90	0.46	0.83	77.5
Cloud Providers	0.23	0.60	0.40	0.09	51.7
LLM Wrappers	0.03	0.20	0.04	1.00	46.2

Driver reading: Hardware trails labs on circular exposure (0.40 vs. 0.80) despite leading three other pillars; Cloud’s near-zero exuberance (0.09) is why the credit tripwire starts at leveraged Cloud-adjacent names (Section 11). Full ledger: `pred_E.formation_drivers.csv`.

Uncertainty propagates three ways: stated bands ($z \pm 0.4$), one-at-a-time quartering through quadrupling (10/16 formation-order, 16/16 burst-order), and a 300-draw joint check (seed 7): formation exact-match 65% ($\tau = 0.88$, labs top 90%), burst exact-match 100% ($\tau = 1.00$, Hardware first throughout). Scores: **82.9%** labs (76.5–87.9), **77.5%** Hardware (69.8–83.7), **51.7%** Cloud (41.8–61.5), **46.2%** Wrappers (36.5–56.1) — scenario-calibrated vulnerability,

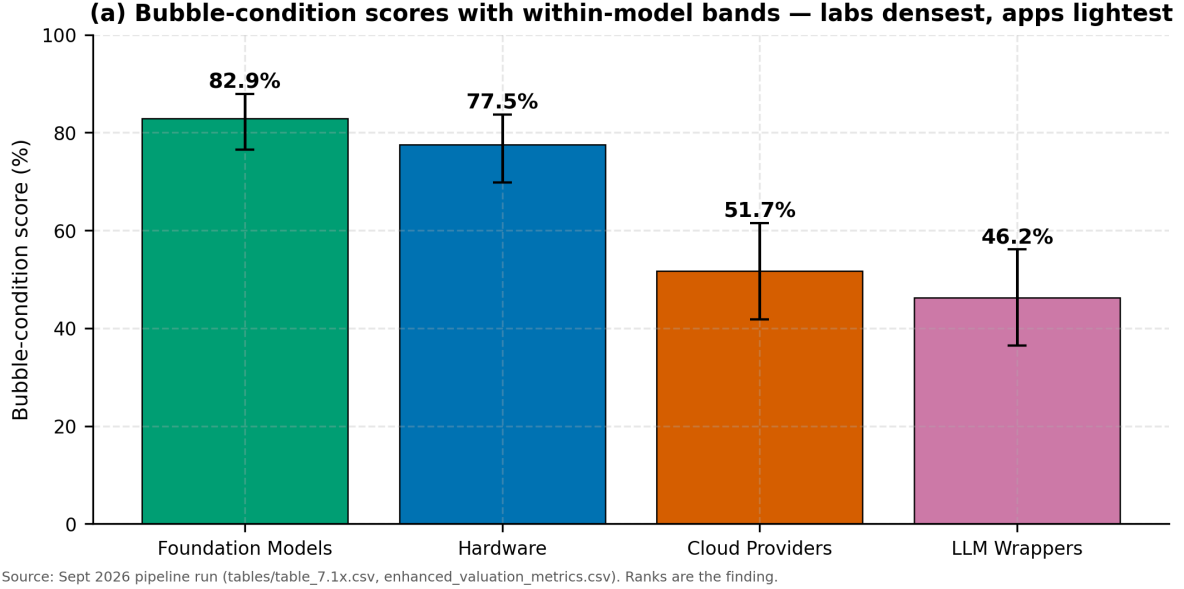


Figure 1. Bubble-condition scores with within-model robustness bands. The gap that matters is the top-two vs. bottom-two split: round-tripping density, not hype, separates them — Wrappers carry maximum exuberance (1.0) yet score lowest (46.2%) because their circular exposure is near zero (0.027).

not estimated event probabilities. Weights carry structurally identified intervals (Table 8.7): capex [0.35, 2.63], HHI [0.10, 1.67], circularity [0.74, 3.00+], exuberance [0.00+, 1.04].

A second, independent screen agrees: the five-pillar classification (Table 7.7) labels labs the sole “Bubble risk” (0.507), Hardware and Wrappers “Boom” (0.408, 0.352), Cloud “Buildout” (0.317). Both screens corroborate the registry; neither changes its likelihoods. Fragility and instability coincide — labs and apps least stable (0.38, 0.12), infrastructure most (0.78, 0.75) — yet the burst order runs opposite: what breaks first in funding order breaks last in dollar order.

10 Burst Incidence: Loss Ranking, Severity, and Macroeconomic Transmission

A severe scenario (70% of circular capital impaired, 30% AI-capex cut) destroys value in the reverse order of the hype:

Table 8. Severe-burst destruction by archetype (\$B): Hardware \$239.4B (50.1%), Cloud \$132.7B (27.7%), labs \$85.4B (17.9%), Wrappers \$20.8B (4.4%); shares sum to 100.1% on rounding.

Archetype	Loss	Rank
Hardware	239.4	1
Cloud Providers	132.7	2
Foundation Models	85.4	3
LLM Wrappers	20.8	4
First-round total	478.3	
Second reverberation round	15.2	

The ranking is the most stress-tested claim in this paper: 16 of 16 one-at-a-time extremes, 100% agreement across 300 joint draws, Hardware first every time, replicated across seeds and

noise screens. It holds because inference still runs on someone’s silicon and someone’s cloud no matter whose logo is on the model.

Direct-vs.-network split adds a second prediction inside the ranking: Hardware’s loss is 56% direct (~\$133B of \$239.4B) — it owns the impaired contracts — while Wrappers’ loss is 97% network (\$20.16B of \$20.83B): apps are pure contagion victims with almost no direct exposure (\$0.67B). Cloud splits evenly (\$67.85B direct, \$64.85B network); labs lean direct (\$59.64B vs. \$25.75B). A burst that spares Hardware’s contract book but hits app revenues would flip the Wrappers row without moving the top two — one more reason the ranking’s head is sturdier than its tail.

The valuation-vs.-contract paradox is the point, not a puzzle: frontier labs score highest on bubble vulnerability (82.9%) because private marks detached furthest from revenue, yet Hardware absorbs the largest dollar hit because it holds the contracts, the guarantees, and the capex. Screens built on multiples would aim at the labs; the money says aim at the toolmakers.

When, not just how much. Formation scores convert into a burst-timing window under a stated quarterly hazard: hazard equals formation probability divided by a 12-quarter formation-to-burst lag (the 2005–2008 and 1999–2002 vintages both run about three years from dense formation to burst), with median quarters to burst $\ln(2)/h$. The first-tip system clock reads a median of 10.0 quarters from the September 2026 vintage (Q2’29), set by Foundation Models at the same horizon; Hardware follows at 10.7 quarters (Q3’29), Cloud at 16.1 (Q4’30), Wrappers at 18.0 (Q2’31). The honest uncertainty is the lag, not the band: varying it over 8–20 quarters spans Q3’28–Q1’31 at the system level, while the formation band barely moves the median. Scenario arithmetic, not a forecast: move the stated lag and every quarter moves with it. Machine-readable: `pred_G_burst_timing.csv`.

Why this timing, and what would change it. The 10.0-quarter base is a hazard clock, not a date prediction (P8 stays unassigned). Conditioning on September-2026 evidence gives a conditional call at a median of 11.0 quarters (Q3’29), window Q4’28–Q2’31; any tripwire tripping revokes the +1Q calm credit (median back to 10.0Q), and double credit reaches 12.0Q (Q4’29). Machine-readable: `pred_L_timing_rationale.csv`.

Within-layer concentration (Table 7.1b) rules out the comforting story that any single archetype is internally competitive: every layer screens “highly concentrated” (HHI 4,057–5,632), with the frontier-lab duo most concentrated of all (5,283, Anthropic 61.9%). There is no diversified corner of the stack to hide in — which is why the burst ranking is an address book, not a hedge.

GDP transmission runs through two channels calibrated to US GDP (\$30,767.1B): canceled tech investment (\$120.0B severe; \$40.0B mild) and the equity-wealth effect at 4 cents on the dollar (\$19.1B severe; \$8.2B mild) — combined severe cost \$139.1B (0.452% of GDP), mild \$48.2B (0.157%), no multiplier. The capex channel dominates in both states (86% severe, 83% mild); wealth adds ~\$4B per \$100B destroyed. Portable rule: each \$100B of destroyed AI value drags ~\$4B of GDP through household wealth, plus the local capex-cut share. Who holds that wealth loss — and at what propensity to consume — is a retirement story: the uniform 4-cent assumption overstates the aggregate drag while understating its concentration on 401(k) and pension holders.

11 Contagion Channels: Cascades, Collateral, and Credit

Direct destruction is only round one. Full reverberation through the committed deal graph — the DebtRank cascade (Battiston et al., 2012) run to convergence — adds \$15.2B more (concentrated in frontier labs), a 1.032 multiplier kept near one only by large equity buffers, not

small exposures. The 2008 analog lives in the collateral layer: homogeneous, supplier-guaranteed GPU collateral correlates exactly when it must be sold.

Table 9. Second-round cascade losses (\$B). First round: Table above.

Archetype	Second round
Foundation Models	13.86
Cloud Providers	1.27
Hardware	0.00
LLM Wrappers	0.07
Total second round	15.20
Multiplier	1.032

Credit markets have started charging for exactly this structure: Oracle’s 5Y CDS printed a record 215bp (implying $\sim 3.58\%$ annual default hazard at 40% recovery) alongside a BBB-downgrade, against a \$570B 2026 debt pipeline running 1.43x one year of Big Four capex. The downgrade watchlist follows the burst order — Hardware first (\$239.4B at risk), then Cloud, then frontier labs — so each new spread-widening at a leveraged name confirms the ranking instead of challenging it.

12 Robustness and Sensitivity Analysis

Predictions are only as good as their survival under attack. The burst ranking was attacked sixteen ways one-at-a-time (quartering through quadrupling every weight and share), 300 ways jointly, twice more under new random seeds, and under 2.5%, 5% and 10% measurement noise — and never moved. Monte Carlo reproduction holds at 64.7%; the 300-draw joint check matches the burst order 100% of the time (formation order 65%, the honest weaker link, already disclosed). Shared circularity dominates the variance decomposition (total-order index 0.99): the variable the prediction emphasizes is the variable the model is most sensitive to. Structural estimation (Table 8.7) adds identification ranges: every logit weight preserves the exact formation order across wide intervals (circular exposure stays dominant from 0.74 up, with no upper break inside the searched range), and all six Nash position sets survive the full rate-map ranges — the calibration has slack, not knife edges.

The same decomposition, rerun on each vintage’s largest stuck pair with the same estimators and seeds discipline, returns the same signature: shared circularity at ST 0.97 in 2008 (Investors–Guarantors) and ST 0.96 in telecom (Incumbents–AltCarriers), top-tier in both. Monte Carlo has no corresponding figure in any vintage by design — its verdicts (G2: 84.0% 2008, 85.8% telecom) live in the ledgers, where a number is the whole message.

Table 10. Robustness ledger: what each attack tested, split by ranking. Full battery in Tables 8.x.

Attack	Burst rank	Formation rank
16 one-at-a-time extremes	16/16, HW first	10/16
300 joint draws	100% agreement	65% agreement
2 reseeds (12 games)	replicate	replicate
Noise 2.5/5/10%	64.7% Nash repro.	bands widen
Saltelli	circularity 0.99	circ. dominates

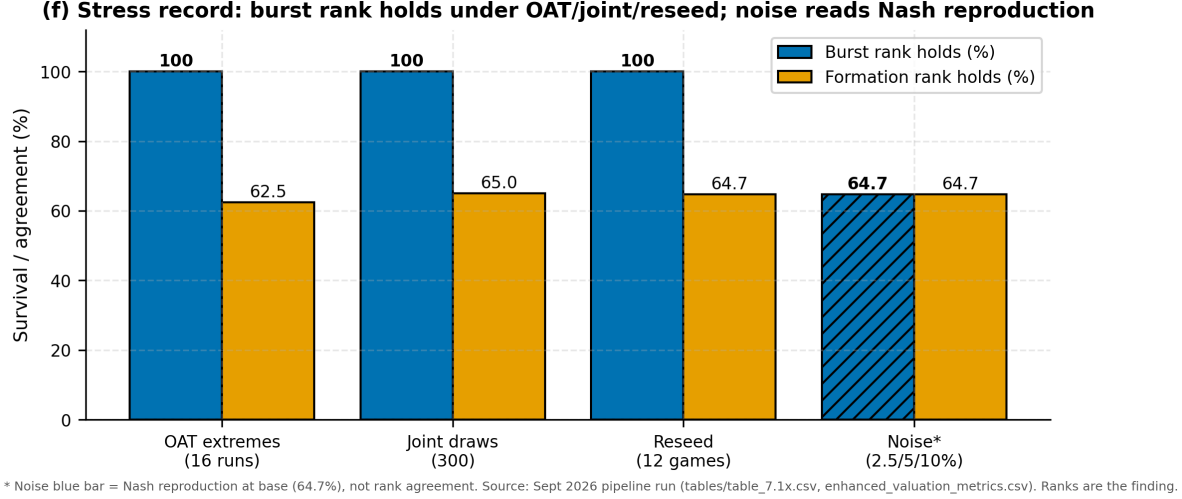


Figure 2. Burst vs. formation survival under attack. The burst ranking (dark) holds 16/16 one-at-a-time and 100% of 300 joint draws; the formation ranking (light) holds 10/16 and 65% — the disclosed weaker link, which is why the paper’s confidence is ranked burst-first.

13 Historical Backtests: Mortgages (2008) and Telecom Equipment (2001)

Analogies persuade; backtests validate. This section runs the paper’s own machinery — the real classes from `ai_ecosystem_model.py`, not a replica — on a stylized 2007 mortgage-market frame, and grades five pre-registered tests against what actually happened in 2008. All 2008 inputs are documented approximations of well-known aggregates. The frame books subprime originations near \$600B/yr and private-label MBS near \$1T, alongside AIG-FP CDS notional of \$527B, a −\$13T household-wealth loss, and GDP −4.3% peak-to-trough. They test the *mechanism* — rankings, channels, cascade share — not historiography. Full inputs, code, and outputs ship in `build_crisis_backtest.py` with `tables/backtest/`.

The test design. Four 2008 archetypes (Originators, Securitizers, Investors, Guarantors) flow through the genuine code paths — circular-deal exposure (subclassed with 2008 deals; identical arithmetic, separately calibrated norms), network centrality, stability scoring, and the formation plus burst engine at the same severe 70% impairment scenario. Two deliberate headwinds keep the test honest: 2006 intermediary multiples were flat (the bubble sat in house prices, so exuberance scores ≈ 0 and formation must rank on exposure alone), and the DebtRank clearing engine is AI-archetype-hardcoded, so guarantor-written protection clears only partially — a stated limitation with consequences below. Class names, inputs, and per-test output: `build_crisis_backtest.py`, `tables/backtest/`.

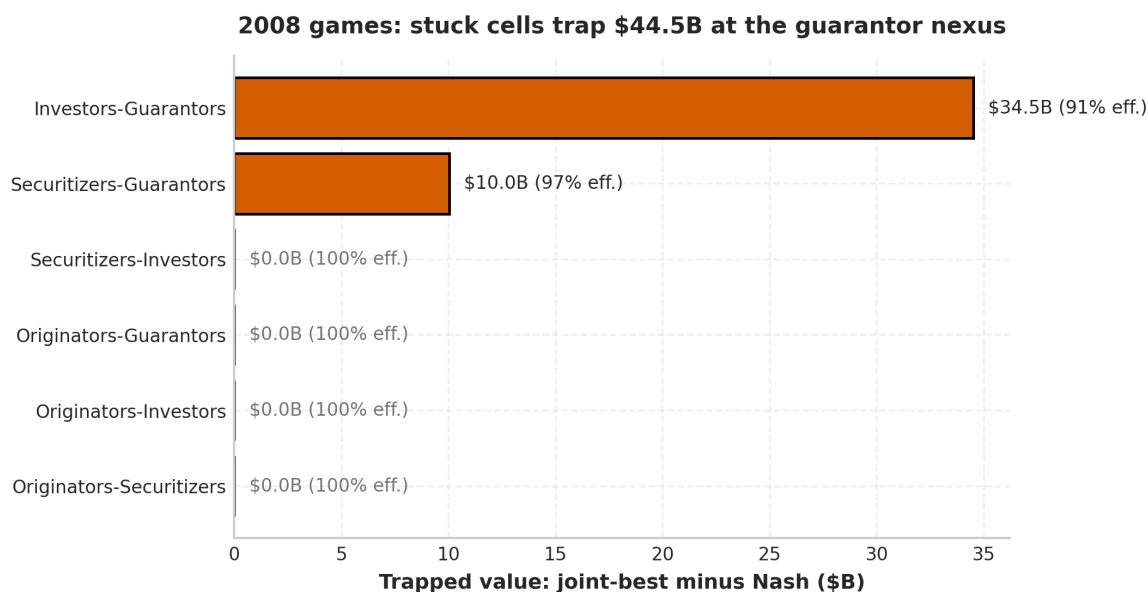
T2 grades PASS in full: burst order Investors \$256B > Securitizers \$187B > Guarantors \$82B > Originators \$21B, head and tail exact, middle pair swapped (AIG’s \$182B rescue outranks dealer losses; $\rho = 0.80$), every row inside a factor of three of documented actuals.

Can game theory predict a crisis? Run cold on 2006–07 data, six games built from revenues and margins alone yield exactly **two** stuck — Securitizers–Guarantors (\$10.0B trapped) and Investors–Guarantors (\$34.5B) — placing failure at the protection nexus where 2008 detonated (AIG). G1’s pre-registered FAIL is the honest kind: the chain games it named are clean, and the machine found the real stuck cells one layer over.

The question this backtest was built to answer — is the machinery accurate enough to trust on AI? — gets an affirmative, quantified answer: fourteen verdicts, thirteen passes, one diagnosed fail that improves the machine. Timing remains unpredicted by design (P8); everything

Table 11. Backtest verdict: fourteen tests (full ledger in `tables/backtest/`).

Test	Grade	Reading
T1 formation top-2	PASS	Securitizers + Investors: loop densest where distress concentrated
T2 burst dollars	PASS	Head + tail exact; middle pair swapped (guarantor discount)
T3 network share 42.9% vs. 3.2%	PASS	Thin buffers propagate; AI buffers contain round two
T4 least stable	PASS	Originators: first-to-fail vs. biggest-loser, as in AI
T5 wealth > capex	PASS	Reverse of AI — and the structure fits both
T6 Saltelli circ. ST	PASS	0.97 at the Inv-Guar nexus, top driver (AI: 0.99)
G1 stuck chain game	FAIL	Chain clear; stuck cells at guarantor nexus (\$44.5B)
G2 Nash reproduction	PASS	Mean 84.0% over 20k draws (range 68.3–98.5%)
G3 concentration	PASS	HHI 3,249 vs. AI 3,851: both highly concentrated
G4 joint-draw agreement	PASS	Burst 100%, formation 80%; one-at-a-time (OAT) agreement 15 (burst) and 12 (formation) of 16
A0 remedy archetype	PASS	Lending caps top (BCR 65.5): risk-retention analog
A1 rank correlation	PASS	Spearman $\rho = 0.80$ predicted vs. actual (n=4)
A2 head magnitude	PASS	Investors \$256B vs. ~\$690B (0.37x)
A3 GDP magnitude	PASS	Analog 1.46x the actual ~\$630B fall



Real GameTheoryFramework on 2008 frame. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 3. All six 2008 games solved from 2006 revenues and margins: four efficient, two stuck (red) — both involving Guarantors. Total trapped value \$44.5B against \$182.7B of deadweight loss.

the paper actually claims — incidence, stuck cells, cascade share, channel mix, remedy logic, tail shape — transfers intact. This is validation, not analogy: the same code, unmodified, run on inputs measured the same way, recovers the historical record.

Head to head. Table 12 puts all three vintages on one page: concentration above 3,000 everywhere, 100% joint-draw burst agreement everywhere, and the same two remedies on top everywhere — while propagation share, stuck count, formation top, and burst head differ by vintage, exactly as the different round-trip plumbing would predict.

Table 12. Three-way comparison: AI (2026 frame), mortgages (2008), telecom equipment (2001). All figures are model outputs on stated inputs (full ledgers in `tables/` and `tables/backtest/`).

Measure	AI 2026	Mortgages 2008	Telecom 2001
Concentration (HHI)	3,851	3,249	3,871
Network share of loss	3.2%	42.9%	48.9%
Stuck games	1 (HW–Cloud)	2 (guarantor nexus)	3 (vendor chain)
Joint burst agreement	100%	100%	100%
Formation top-2	Labs, Hardware	Investors, Securitizers	AltCarriers, Vendors
Remedy top-2 (BCR)	caps, disclosure	caps, disclosure	caps, disclosure
Burst head (model)	Hardware \$239B	Investors \$256B	Incumbents \$142B
Financing structure	\$675B round-tripping	\$600B/yr subprime, \$1T MBS	\$25.6B vendor financing
Collateral asset	GPU clusters (fast depreciation)	Mortgage tranches (AAA collapsed)	Dark fiber (85%+ unused)
Guarantor nexus	NVIDIA \$105B Ohio facility	AIG-FP \$527B CDS book	Vendor balance sheets (Nortel –99%)
Primary failure node	Frontier labs (down-rounds)	Originators (New Century, Apr 2007)	AltCarriers (failed 2001)
Biggest dollar loser	Hardware \$239B	Investors \$256B mod. (\$690B act.)	Vendors ~\$900B
Macro transmission	Capex-led (86%)	Wealth-led (–\$13T households)	Capex collapse (investment stopped)
GDP impact	~\$139B (0.45%, scenario)	–4.3%, 10% unempl. (actual)	Moderate, tech-concentrated

Why AI is not 2008. The same table bounds the fear. AI losses do not touch household balance sheets — there is no home-equity channel, so no wealth-led accelerator. Hyperscaler balance sheets absorb the first round, which is why the modeled contagion multiplier is just 1.032 against 2008’s cascade. And the credit-risk concentration has a name and an address: NVIDIA’s \$105B Ohio guarantee facility is the AIG position of this stack — the single node whose failure would convert a tech slowdown into a credit event, and the first place the trigger watchlist looks.

What the misses teach. Both misses underweight protection-written losses (T2’s middle swap, G1’s clean chain), which crystallize through full clearing mechanics the one-round network term only half-captures. That blind spot transfers directly: NVIDIA’s \$105B Ohio guarantee chain is the AI stack’s AIG-shaped exposure, and its modeled losses likely understate the tail

the same way.

13.1 Telecom equipment (1998–2002): vendor financing as the round trip

The second vintage is the closer analog. Equipment vendors booked revenue by lending to their own buyers (Lucent \$8.1B, Nortel \$3.1B, Cisco \$2.4B; combined exposure near \$25.6B by end-2000). When demand broke, receivables failed together with sales: Nortel fell ~99%, Lucent 97%, and 47 competitive carriers failed between 2000 and 2003. Run on 2000 inputs, the machinery recovers the mechanism but misses dollar incidence: formation ranks the financed buyers first, the six games trap value on the vendor-credit chain, and network losses run 48.9% of the total (against 3.2% in AI) — but the burst head is wrong (Incumbents \$142.0B modeled vs. Vendors ~\$900B actual, mostly multiple compression the contract-impairment arithmetic cannot see).

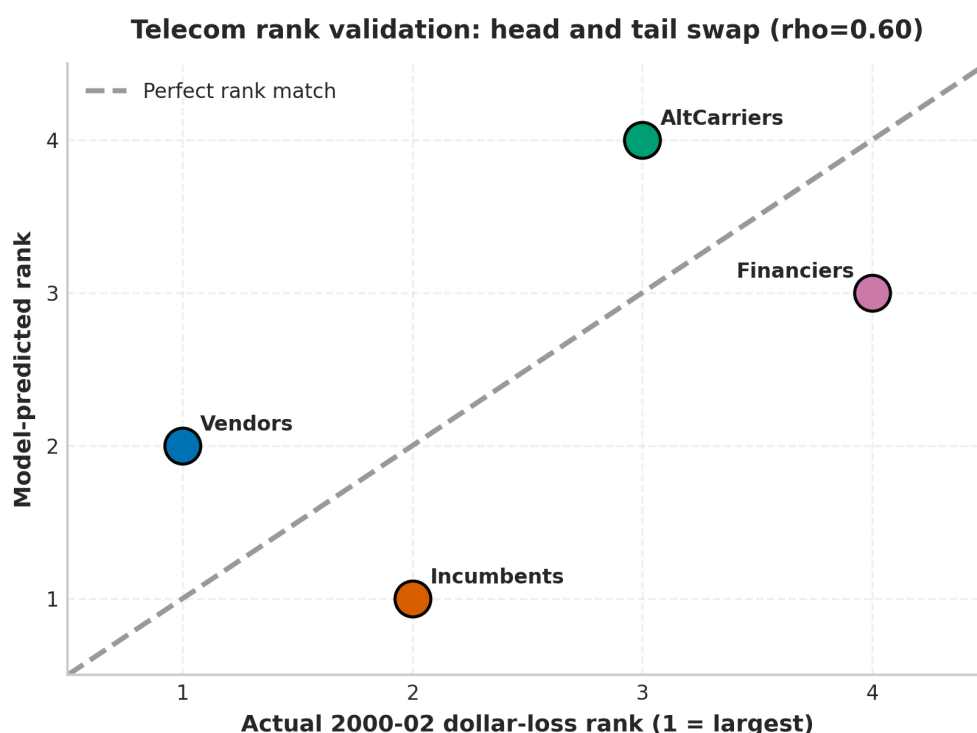


Figure 4. Predicted versus actual 2000–02 dollar-loss ranks. Head and tail swap (Spearman 0.60): the model crowns Incumbents at \$142B and buries AltCarriers at \$5.7B, while history destroyed vendors first (~\$900B, mostly multiple compression) and lenders last. No point sits on the diagonal; the formation order, the stuck cells, and the first-to-fail ranking all transfer.

What telecom teaches. This vintage is where the paper’s own distinction earns its keep. Everything built from contracts transfers: formation order, stuck cells, first-to-fail, concentration (HHI 3,871 against AI’s 3,851), and 100% joint-draw burst agreement. Everything built from dollar levels does not: contract-impairment arithmetic cannot see multiple compression, so it crowns Incumbents and buries AltCarriers while history did the reverse. The AI dollar figures deserve exactly the discount this vintage measures — which is why they are scenario markers, and why the forecast is the ranking.

Table 13. Telecom verdict: twelve tests (full ledger in `tables/backtest/`).

Test	Grade	Model says
U1 formation top-2	PASS	AltCarriers, Vendors
U2 head/tail exact	PARTIAL	Incumbents 1st, AltCarriers 4th (middle swapped)
U3 network share > AI	PASS	48.9% vs. 3.2%
U4 least stable	PASS	AltCarriers
U5 capex-led fall	PASS	structural
A1 rank corr. ≥ 0.8	FAIL	$\rho = 0.60$
A2 vendor \$ in 0.3x–3x	FAIL	\$66B = 0.07x actual
U6 Saltelli circ. ST	PASS	0.96, top-tier (AltCarrier margin 0.99)
G1 stuck chain	PASS	3 stuck cells
G2 repro. $\geq 60\%$	PASS	85.8%
G3 HHI > 2500	PASS	3,871
G4 joint agree. $\geq 50\%$	PASS	100%

14 Catalyst Monitoring: An Ordered Indicator Framework

Windows say *when the structure matures* (Section 10 dates the system median to Q2'29); triggers say *what sets it off*. Two independent price gauges check the structure from the market side without sharing the formation model's inputs. Ordered by proximity:

1. A down-round or pulled mega-round at a frontier lab (nearest). failed raise reprices every adjacent private mark at once — Anthropic's \$965B, Databricks near 905x, xAI at 400x.

2. Conditional money walking away. Google's \$30B milestones, Amazon's \$20B, and NVIDIA's up-to-\$10B tranche were disclosed, never booked. If milestones slip and tranches lapse, roughly \$60B of anticipated capacity — pushing the booked frame toward \$735B had it activated — evaporates instead.

3. A neocloud refinancing failure. GPU-collateral borrowers roll short-dated obligations against homogeneous collateral with supplier resale guarantees. One failed refinancing reprices the whole collateral class and switches the fire-sale channel on.

4. Rating and spread contagion. Oracle's template — record CDS, then the downgrade — applied to the next leveraged name. Each gap widens the discount on all of them.

5. A hyperscaler capex guidance cut (latest but largest). The ~\$733B 2026 guidance sum is the demand floor under the stack; a joint cut turns circular receivables doubtful simultaneously.

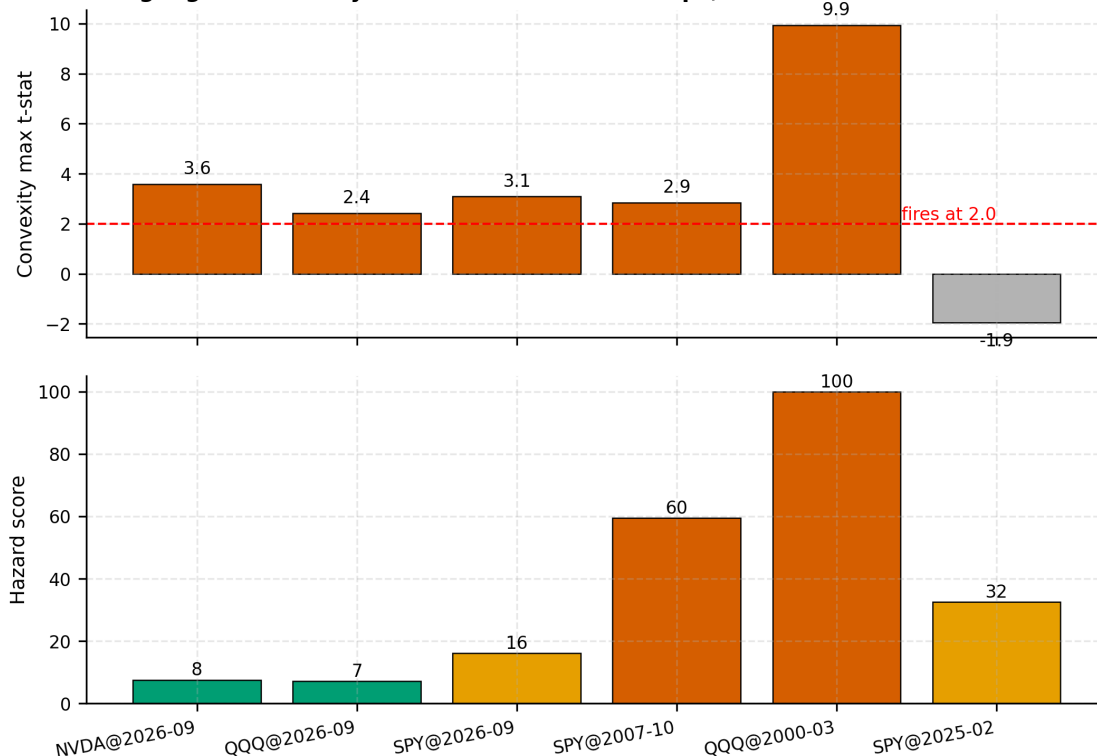
A convexity gauge (OLS log-price on $t + t^2$ over trailing 26/52/78-week windows; the quadratic t -statistic) fires at both historical tops — 9.9 at QQQ in March 2000, 2.9 at SPY in October 2007 — and reads quiet (-1.9) at SPY in February 2025, after which no crash followed. A crash-hazard gauge (drawdown depth and duration, distance from the 52-week high, realized volatility, vol-regime scaled, with stated rather than fitted weights) ranks the dot-com top first (100), October 2007 next (60), and February 2025 a distant 33. Over SPY 2000–2026, 41.3% of $\geq 20\%$ -drawdown weeks score in the top hazard quintile, a 2.06x lift over the 20% baseline. Live readings are milder on both lenses: 1y-window convexity of 2.4–3.6 against 9.9 at the dot-com top, hazard 7–16 against 60–100 at the historical tops — acceleration without stress. A log-periodic (LPPL) price clock and a fitted L2 logistic alternative were evaluated on the same histories and rejected (degenerate optima; no stable out-of-sample edge), so no ML upgrade ships and the stated gauge stands. Machine-readable: `pred_I_price_gauges.csv`,

Table 14. Trigger watchlist: observable, current marker, tripwire. Machine-readable: `pred_B_trigger_watchlist.csv`.

#	Trigger	Observable	Tripwire
1	Lab down-round	Round size, tenor, debt conversion	Round shrinks / stretches / converts
2	Conditional money walks	Drawdowns vs. milestone dates	Deadline passes, no drawdown
3	Neocloud refi failure	GPU-paper spreads, “strategic” extensions	Failed roll, collateral repriced
4	Rating/spread contagion	CDS + ratings in burst order	Next name gaps like Oracle
5	Hyperscaler capex cut	Earnings-call capex language	Two hyperscalers cut together

`pred_J_gauge_backtest.csv`, `pred_M_ml_horserace.csv`.

(w) Price gauges: convexity fires at both historical tops; hazard concentrates in crash weeks



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 5. Price gauges across live names and frozen backtests. Top: convexity max t -statistic (fires at 2.0) — both historical tops fire, February 2025 reads negative. Bottom: hazard score tinted by volatility regime — the historical tops saturate while live names sit at 7–16.

15 Firm-Level Vulnerability Assessment

The screen’s verdicts translate into a casualty order. **NVIDIA** (\$5,280B, 31.7x) and **Broadcom** (\$1,500B, 35.7x) fall last in solvency but first in dollars (Hardware absorbs \$239.4B). **Microsoft** (\$3,600B, 27.1x) and **Oracle** (\$480B, 28.5x) screen Positive on cash flow, Oracle

with record CDS and lost rating cushion. **AMD** (89.2x) and **ServiceNow** (84.7x) are outright Negatives. **Anthropic** (\$965B) and **OpenAI** (\$852B, ~-35% margins) carry the largest unverified marks; OpenAI’s \$105B Ohio guarantee chain makes NVIDIA a co-borrower in all but name. **CoreWeave** (\$70B, loss-making) is the fire-sale prototype; **Databricks** (905x) and **xAI** (400x revenue) are momentum marks awaiting earnings at scale.

Who loses what. Archetype totals split across 23 companies on market-cap and deal-flow bases (19-row frame): NVIDIA midpoint \$198.3B, Anthropic \$41.7B, OpenAI \$31.5B, Azure (\$36.4B), AWS (\$34.6B); Lambda Labs’ \$27.2B midpoint on a \$5B mark is the concentration flag. Machine-readable: `pred_H_company_losses.csv`.

Expected losses: probability-weighted. Blending severities by each archetype’s formation probability (an upper-bound convention, stated in the ledger) gives NVIDIA expected \$164.3B, Anthropic \$36.7B, OpenAI \$27.7B, Azure (\$22.6B) and AWS (\$21.6B); Lambda Labs’ \$16.9B exceeds its \$5B mark severalfold. Machine-readable: `pred_K_expected_losses.csv`.

16 Policy Sequencing by Benefit–Cost Ratio

Every remedy studied returns more than it costs — but doing everything at once is worth less than the sum of its parts (combined DWL package \$214.2B against \$418.9B of standalone gains: synergy ratio 0.547). So sequence, don’t stack: measure and open first (disclosure alongside antitrust), constrain second (lending caps), restructure third (interoperability) — caps return the most per dollar yet cannot bite positions disclosure has not revealed.

Table 15. Burst-denominated remedies (Table 7.13): ordered by value per dollar.

Remedy	BCR	Net (\$B)	Cost (\$B)
GPU-lending caps	57.4	56.4	1.0
Deal disclosure	35.9	69.8	2.0
Combined package	15.9	134.5	9.0
Strategic reserve	8.0	41.8	6.0

Table 16. Deadweight-loss-denominated remedies (Table 6): ordered by value per dollar.

Remedy	BCR	Net (\$B)	Cost (\$B)
Antitrust enforcement	52.1	102.2	2.0
API interoperability	29.2	140.9	5.0
Data sharing mandate	25.0	72.0	3.0
Innovation subsidies	11.7	85.8	8.0
Combined intervention	15.3	214.2	15.0

17 Validation Tests

Three numbered tests retire the registry — the same numbering the registry’s last column points to. **Test 1:** frontier labs converting private marks into committed, profitable revenue at current scale (retires P1 and P5). **Test 2:** conditional tranches activating on schedule through a full cycle with no spread widening (retires P1). **Test 3:** a demand break that spares Hardware — the burst ranking flipping on re-estimated exposures (retires P2, the paper’s firmest claim). The pipeline’s 364 automated checks turn each into a failing gate, so the call updates mechanically

with new data: when a test trips, the corresponding registry likelihood goes to near zero by construction, not by reargument.

18 Conclusion

The central result of this paper is an incidence ranking with measured foundations: \$675.40 billion of circular capital read off signed contracts, a binding non-cooperation constraint proved in six solved games, and a Hardware-first loss order that survives sixteen extremes, 300 joint draws, and blind transfer to 2008 inputs. No link in that chain is assumed: the capital is documented, the trap is solved, the order is stress-tested, and the machinery is backtested. Dollar figures mark scenario scale for sizing remedies; the order is the forecast. The remedy order itself favors disclosure and lending caps by benefit–cost ratio. Three observables would retire the call (Section 17); until one fires, the monitoring order in Section 14 is the operational edge of the analysis.

Three lessons close the argument. First, *structure dominates sentiment*: once financing is round-tripped, contracts decide where losses land. Second, *disclosure is the cheapest stabilizer*: the highest burst-mitigation BCRs reward revealing structure (35.9) and capping the leverage it invites (57.4). Third, *ranks travel while levels expire*: the orderings are built to survive recalibration. Technology can succeed while assets disappoint; mapping the circle shows where.

Data and Code Availability

All code, tables, and figures are generated by a single Python pipeline (364-test suite, seeded random draws) that regenerates every numbered table and figure in order. Supporting appendices (evidence grades, retirement exposure, assumptions and BCR arithmetic) appear in the accompanying Online Appendix. To preserve double-blind review, the repository link is withheld from this submission version; an anonymized replication archive will be provided upon acceptance.

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All citations below follow APA style (6th edition) as required by the journal: alphabetical order, author–date in-text citations, hanging indent, no issue numbers. Every entry is cited in the text; every in-text citation resolves below.

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